Selecting a data product requires comparing what the product is designed to provide with the needs of the air quality analysis. A large geographic coverage can be helpful, but it does not guarantee that the data level or product design is appropriate for a specific air quality application. Visualization colors do not determine whether a product is scientifically appropriate for an air quality task. Recency alone does not determine whether a product fits the air quality question being asked.

True-color imagery captures visible wavelengths, allowing dense aerosol plumes to be seen while trace gases remain undetectable. True-color imagery depends on visible light, so it provides little to no observation at night. True-color imagery can suggest plume presence, but it does not directly identify every aerosol type or quantify concentration. The visibility of smoke does not mean trace gases can be detected or measured with true-color data.

Reanalysis uses data assimilation to align a model with observations and produce surface-level air quality estimates. It depends on a model informed by observations rather than a direct conversion from satellite column data to surface concentrations. Reanalysis uses data assimilation to analyze past events while forecasts predict future conditions. Creating finer-scale pollution maps is a common purpose of hybrid or fusion products.

Satellites can be used to track changes in air quality on a regional scale in near-real-time, such as when lockdowns during the COVID-19 pandemic led to reduced air pollutant emissions in the Eastern U.S. in 2020.

The global vantage point and long-term continuity of satellite data records can allow us to track changes in air quality due to national pollution control policies, such as China's efforts to reduce industrial pollution.

Satellite data allows tracking and early-warning of pollution which might be coming from far away sources, such as dust being blown across the Atlantic Ocean.

Satellites provide information to help fill in gaps between ground-based air quality monitors, and can support air quality advisories in remote areas.

Satellites provide the globally consistent data needed for worldwide studies of the health impacts of air pollution, such as the 2023 Global Burden of Disease study.

Satellite remote sensing can cover large and remote areas, which is a significant advantage over ground-based monitors.

Satellites offer a global perspective, enable long-term pollutant tracking, and assist in evaluating pollution control policies.

Activity Module 4: Search and Select Data Products

Completion requirements
To do: Receive a grade

Task 1: Compare AOD from Different Satellites

Use the embedded Worldview window below or open the **Worldview Task 1 link** (opens in a new browser tab). Using the Corrected Reflectance (True Color) from Suomi NPP / VIIRS base layer, locate southern Afghanistan (location 1). Notice how the land surface is fairly uniform and bright compared to surrounding land features. Recall from the Module 3 activity we said the dominant aerosol here is dust, and this region is a combination of several deserts in Afghanistan.
Click the "**Add Layers**" button, then use the search bar at the top of the window to search for and add the product layers listed in the table. Note that two of the layers are the orbit track & time for the Aqua and Terra satellites.

Layer Name	**Platform/ Sensor**
Deep Blue Aerosol Optical Depth (Land)	Aqua / MODIS
Deep Blue Aerosol Optical Depth (Land)	Terra / MODIS
Aqua Orbit Track & Time Ascending Day	Aqua / Space-Track.org
Terra Orbit Track & Time Descending Day	Terra / Space-Track.org

Examine the AOD products from the Deep Blue algorithm for MODIS on both satellites over southern Afghanistan (location 1). Hide and show each layer to compare AOD values.

Why do MODIS/Terra and MODIS/Aqua AOD, derived from the same algorithm, differ in southern Afghanistan (location 1)?
Your answer :

The times of the two orbit tracks (i.e., satellite overpass timing) differ with one morning and the other afternoon.
Response: Correct: Satellite overpass timing differs. Terra overpasses in the morning and Aqua in early afternoon. Changing weather in that ~4-hour window, like stronger afternoon winds potentially kicking up dust west of location 3, yields higher Aqua AOD values.

Task 2: Compare AOD Products Over a Bright Desert Region in Southern Afghanistan
Use the embedded Worldview window below or open the **Worldview Task 2 link** (opens in a new browser window) to start with a clean map, or remove the AOD and orbit layers from the previous task, except the true color base layer, by clicking the "x" in the upper right of each layer box. Zoom into southern Afghanistan (location 1).

Search for and add the product layers listed below. Examine the desert region of southern Afghanistan as in the prior task. Note there are four AOD products listed, all distinct with respect to retrieval algorithm, satellite, and sensor.

Layer Name	**Platform/Sensor**
Merged DT/DB Aerosol Optical Depth (Land and Ocean)	Aqua / MODIS
Merged DT/DB Aerosol Optical Depth (Land and Ocean)	Terra / MODIS
MAIAC Aerosol Optical Depth	Terra and Aqua / MODIS
Dark Target Aerosol Optical Depth (Land and Ocean)	NOAA-21 / VIIRS

Task 2: Summary

- Dark Target AOD data from VIIRS is missing over southern Afghanistan (location 1), even though it does exist over other land and water areas nearby. As described in the Module 4 lesson, the Dark Target AOD product performs poorly over bright surfaces like desert. The Dark Target algorithm will also mask out clouds, but there are no clouds in the true color imagery.
- The MAIAC AOD combines information from two sensors - Terra and Aqua MODIS, while the Merged Dark Target (DT) / Deep Blue (DB) AOD products merge information from these two algorithms from a single sensor. However, we cannot state that one of these products is more accurate than the other based on the available information.
- The option stating that there are notable differences between the MODIS Merged DT/DB products is correct. A likely reason for the difference is the same as in Task 1 - the products are from different sensors that have different overpass times, MODIS/Terra in the morning and MODIS/Aqua in the afternoon.
- The option stating that each product is similar and can be used interchangeably is incorrect because by showing and hiding each product you can see clear differences. These differences arise due to many reasons covered here and in Module 4: different sensors, algorithm choice, and time of day the observations were made.

Task 3: Selecting an AOD Product in the Presence of Dense Smoke in Northern India and Eastern Pakistan

Use the embedded Worldview window below or open the **Worldview Task 3 link** (opens in a new browser window) which includes the same layers as Task 2.
Add the map marker for this location by entering the coordinates into the search box for places in the upper right corner of the map: 30.8290°, 72.1432°.
Compare AOD products over the northern India and eastern Pakistan region.

Layer Name	**Platform/ Sensor**
Merged DT/DB Aerosol Optical Depth (Land and Ocean)	Aqua / MODIS
Merged DT/DB Aerosol Optical Depth (Land and Ocean)	Terra / MODIS
MAIAC Aerosol Optical Depth	Aqua & Terra / MODIS
Dark Target Aerosol Optical Depth (Land and Ocean)	NOAA-21 / VIIRS

Recall from the Module 3 activity that smoke particles were the dominant aerosol at this location. There are substantial gaps in AOD data above the location from all products listed here, except for MAIAC.

What is the most likely reason for the missing AOD from the Merged DT/DB and Dark Target AOD products in northern India and eastern Pakistan (location 2)?

Your answer :

Dense smoke is mistaken for clouds and the AOD retrievals are masked out.
Response: Correct: Dense smoke is mistaken for clouds and the AOD retrievals are masked out.

Task 3 Summary

- Dense smoke can be highly reflective and appear to be clouds. AOD algorithms assign quality flags to let users know to potentially filter out the value at that point, e.g., due to cloud cover. Worldview uses those flags to mask AOD values in areas of clouds.
- The higher spatial resolution of MAIAC compared to Dark Target and Deep Blue provides more opportunities for the retrieval to “see around” small clouds, but the wavelengths it uses can not “see through the smoke.”
- While the Dark Target and Merged AOD products are optimized for specific surface types, their algorithms aren't explicitly tuned for episodic events like wildfires.
- Although data is missing over this location, the data for surrounding points suggests that the AOD detections are being masked out as opposed to no satellite overpass having occurred there.

As seen in Task 3, most AOD products mask out the thickest parts of smoke plumes. When this occurs, we can use the Absorbing Aerosol Index (sometimes called 'Aerosol Index' (AI) or Ultraviolet AI (UVAI)) to observe highly absorbing smoke aerosols.

UV Aerosol Index is used to account for anything besides ozone that absorbs UV light in the atmosphere so the algorithm can calculate ozone accurately. Other products from TEMPO’s ozone processing, such as “Tropospheric Ozone Column”, are also going to appear in the parentheses of the layer title. **Note, TEMPO is a GEO instrument so zoom out to see where the TEMPO “Ozone” UVAI data exists.**

If you want to study the thick portion of this smoke plume over northern India and eastern Pakistan (location 2), which instrument(s) and product is most appropriate?

Your answer :

Aura OMI UVAI
Response: Incorrect: Suomi NPP's OMPS AI is correct because its spatial resolution and coverage outperform the UVAI Aura OMI for identifying smoke, and TEMPO covers North America, not India.

Task 4 Summary

- TEMPO is on a geostationary orbit fixed over North America, thus its field of regard does not cover India. It is worth noting that similar products are available from the Geostationary Environment Monitoring Spectrometer (GEMS) instrument on South Korea's GEO-KOMPSAT-2B satellite, but these data are not currently available in Worldview.
- UVAI from Aura OMI shows the coarse spatial resolution of the data. The AI from Suomi NPP OMPS has higher spatial resolution and no data gaps, making it the best choice for daily smoke monitoring over northern India and eastern Pakistan (location 2).

Task 5: Selecting an Aerosol Index (AI) Product for Long-term Studies in Northern India and Eastern Pakistan

The previous task examined AI data products for a single day (November 8, 2024). Let's consider how to examine the average occurrence of thick smoke in northern India and eastern Pakistan (location 2) for the same month (e.g., July) over the last 20 years. Which AI product is best suited for this question?

If we wanted to study trends over a 20-year period, we'd first have to find which data would be available at the start of that period.

Layer Name	**Platform / Sensor**
Aerosol Index	Suomi NPP / OMPS
UV Aerosol Index	Aura / OMI
Ozone (L3, UV Aerosol Index, Subdaily)	TEMPO

Which statement accurately describes the availability of absorbing AI products between 2004 and 2024?

Your answer :

OMI UVAI is available for 2004 and OMPS AI starts in January 2012.
Response: Correct: OMI, launched in 2004, covers the full 20-year span. OMPS didn't produce data until 2012, so it can't cover the full period alone. TEMPO data does not cover this region.

Task 5 Summary

Data from the OMI instrument is available for this entire 20 year period. Suomi NPP OMPS did not start producing usable data until 2012. While OMPS data might be useable in the study, it would not cover the entire 20 year period. Again, because the TEMPO sensor is in geostationary orbit over the Americas, data are not available over this region of the world. The field of regard for the GEMS mission mentioned previously covers northern India and eastern Pakistan (location 2); however, the GEMS mission did not launch until 2020 and the data are not available in Worldview.

Task 6: Selecting a Product for Point Source Identification for Central India

Location: central India (location 4, Duddhi, Uttar Pradesh)

In this task, we will view trace gas data for Central India (location 4). As discussed in Module 3 NO2 has a short lifetime, and shows larger values around sources. Thus, NO2 observations are useful for distinguishing individual industrial point sources. In the Module 3 activity, we determined that on November 8, 2024, central India (location 4) was dominated by industrial pollution, specifically from large thermal power plants.

Task 6 - Which Product is More Appropriate to Determine Point Sources for Central India

Which NO2 product most clearly identifies industrial point sources of pollution in central India (location 4)?

Your answer :

TROPOMI (Sentinel-5P), because it has higher spatial resolution.
Response:

Correct: Pinpointing industrial sources requires fine (high) spatial resolution.

Task 6 Summary

Identifying specific point sources like power plants requires high spatial resolution. TROPOMI provides significantly higher spatial resolution compared to the older OMI instrument. While TEMPO provides excellent spatial resolution and the added benefit of hourly observations, it is a geostationary instrument that only covers North America. Other geostationary instruments like GEMS are available for this region but not through Worldview.

Module 4 Activity Summary

This activity guided you through choosing between different data products relevant to air quality in NASA Worldview. For each geophysical parameter (e.g., AOD or tropospheric NO2), there are often multiple products available from different instruments or retrieval algorithms. When choosing between data products, it is important to consider factors like spatial resolution, spatial and temporal coverage, overpass times, and algorithm limitations such as sensitivity of cloud cover or dense smoke. Tools like Worldview can help to quickly compare different data products to select one which provides the best information for a given region, time period, and pollutant or topic of interest.